



AUGSD
First Semester 2026-2027
COURSE HANDOUT (PART II)

In addition to part-I (General Handout for all courses appended to the timetable) this portion gives further specific details regarding the course.

Course No. : CS F429
Course Title : Natural Language Processing
Instructor-in-Charge : Poonam Goyal (poonam@pilani.bits-pilani.ac.in)
Lab Instructor : Sunil Yadav (p20210403@pilani.bits-pilani.ac.in)

1. Scope and Objectives

The intent of the course is to present a broad undergraduate/post-graduate level introduction to Natural Language Processing (NLP, a.k.a. computational linguistics), the study of computing systems that can process, understand, or communicate in human language. The primary focus of the course will be on understanding various NLP tasks as listed on the course syllabus, algorithms for effectively solving these problems, and methods for evaluating their performance.

This subject aims to achieve the following goals:

- To introduce students, the challenges of empirical methods for natural language processing (NLP) applications.
- To introduce basic mathematical models and methods used in NLP applications to formulate computational solutions.
- To provide students with the knowledge on designing procedures for natural language resource annotation and the use of related tools for text analysis and hands-on experience of using such tools.
- To introduce students research and development work in information retrieval, information extraction, and knowledge discovery using different natural language resources.
- To give students opportunities to sharpen their programming skills in neural networks (and deep learning) for computational linguistics applications.

2. Prerequisites

Programming in python and knowledge of core data structures & algorithms and neural networks.

3. Text and Reference Books

Daniel Jurafsky and James H. Martin. 2026. Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition with Language Models, 3rd edition. Online manuscript released January 6, 2026. <https://web.stanford.edu/~jurafsky/slp3> .

R1: Manning and Schütze, Foundations of Statistical Natural Language Processing, MIT Press. Cambridge, MA: May 1999.

R3: Natural Language Toolkit. Bird and Loper, and other developers. Available for free at: – <http://www.nltk.org/>





4. Course Plan

Lecture No	Learning Outcomes	Topics to be covered	Chapter in the Text Book
1-2	To Introduce NLP and its applications, text preprocessing	Introduction to NLP, Text Preprocessing, Standardization	Ch1 & 2
3-5	To apply N-gram models for document generation	N-Gram Language Models, Evaluation, Smoothing Interpolation, etc.	Ch3
6-10	To identify the sentiment of the given text	Text classification, Traditional sentiment analysis, Naïve Bayes	Appendix B and class notes
11-13	To understand how to apply LR to text	Logistic Regression, Sigmoid, Softmax, Cross entropy loss function and Stochastic Gradient Decent algorithm	Ch4
14-17	To convert words into various form of vectors	Word embeddings- co-occurrence matrix based, word2vec (Skip gram and CBOW) Glove, Vector semantics	Ch5
Self-study	To learn Neural Network	Neural Networks and Neural language models	Ch6
18-23	To apply POS tagging on a given corpus	Sequence Labeling- Part-of-Speech Tagging and Named Entity Analysis, hand crafted features	Ch17
24 & self study	To solve sequence labeling through NNs	RNNs and LSTMs	Ch13
24-25	To understand fundamentals of LLMs	LLM Introduction	Ch7
26-27	To understand how LLMs generate text conditioned on a prompt	Conditional Text Generation, Prompting, sampling, decoding	Ch7
28	To learn how models can be trained without large-scale labelled data	Self-supervised pretraining, finetuning, evaluation	Ch7
29-30	To learn the architecture for building LLMs	Transformers	Ch8
31-32	To understand how supervised instruction–response data adapts LLMs to follow user instructions.	Post-training: Instruction Tuning, Alignment, and Test-Time Compute	Ch9
33-34	To learn about bidirectional encoders	Masked Language Models	Ch10



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35-36	To understand how LLMs fulfill the human information needs	Information Retrieval and Retrieval Augmented Generation	Ch11
37-39	To apply Machine Translation on a given parallel corpus and measure the performance of the translation	Machine Translation, Encoder Decoder models, Evaluation	Ch12
40-42	To apply all the NLP related techniques for solving applications and to understand LLMs	Applications: Information Extraction, Question Answering/ Dialog Systems/Chatbots Large language models (LLMs)	Ch20 & 21 Class Notes

5. Evaluation Scheme

Component	Duration	Weightage	Date &Time	Mode
Labs/Assignments	-	35%	To be announced	Take home
Quiz	-	5%	In First month of the Semester	Closed Book
Mid-Term exam	90 mins	25%	16/03 09:00-10:30	Closed Book
Comprehensive	3 hours	35%	15/05 09:00-12:00	Partly Open Book

6. Chamber Consultation: To be declared in the class.

7. Notices: All notices will be put up on Nalanda

8. Make-up Policy: Prior Permission is must and Make-up shall be granted only in genuine cases based on individual's need and circumstances.

9. Academic Honesty and Integrity Policy: Academic honesty and integrity are to be maintained by all the students throughout the semester and no type of academic dishonesty is acceptable.

Instructor-in-charge

